ROADSAFE AI

Intelligent Road Safety & Accident Risk Analysis

FINAL PROJECT PROPOSAL
Machine Learning • Deep Learning • Computer Vision • Natural Language Processing

Project Area	Artificial Intelligence • ML • DL • NLP • Computer Vision
Application	Road Safety & Accident Risk Analysis
Platform	Python + Streamlit
Development Time	7 Days
Project Type	Academic AI Prototype

A focused, one-week implementation designed to demonstrate four major AI disciplines in one interactive application.

1. Introduction

Road accidents are a major safety concern associated with factors such as traffic density, road conditions, weather, visibility, and other environmental circumstances. RoadSafe AI proposes an integrated artificial intelligence prototype that analyzes road images, structured traffic and accident-related data, and written accident descriptions.

The system combines Machine Learning (ML), Deep Learning (DL), Computer Vision (CV), and Natural Language Processing (NLP) in one Streamlit-based application. It is designed as an academic decision-support and research prototype, not as a replacement for professional traffic-control or emergency-response systems.

2. Problem Statement

Road-safety information is often available in different forms, including images, traffic data, environmental information, and textual accident reports. When these sources are analyzed separately, it becomes difficult to obtain a unified view of road conditions. RoadSafe AI addresses this challenge by bringing these sources together in one AI platform.

3. Aim

To develop an AI-based road-safety prototype that uses ML, DL, Computer Vision, and NLP to analyze traffic conditions, process accident-related information, and provide a model-based accident-risk assessment.

4. Objectives

1. Detect vehicles and pedestrians in road images using deep-learning-based object detection.

2. Count detected vehicles and estimate a traffic-density indicator.

3. Train a machine-learning model to classify road-safety risk into Low, Medium, or High categories.

4. Analyze accident descriptions and extract useful information using NLP.

5. Integrate outputs from the AI modules into one interactive dashboard.

6. Evaluate the implemented models using appropriate performance metrics.

5. Proposed System Modules

A. Computer Vision & Deep Learning

Users will upload road images. A YOLO-based object detection model will identify common road objects such as cars, motorcycles, buses, trucks, and pedestrians. Detected objects will be displayed with bounding boxes, and object counts will be calculated.

B. Machine Learning Risk Prediction

A structured dataset will be used to train and compare suitable classification models. Candidate algorithms include Logistic Regression, Random Forest, and XGBoost. Features may include weather, road condition, visibility, traffic density, time of day, road type, and other relevant dataset attributes.

C. NLP Accident Report Analyzer

Users will enter an accident description. NLP processing will extract useful information such as accident type, vehicles involved, weather, road condition, and visibility. The module may use text preprocessing, entity or keyword extraction, and text classification where appropriate.

D. Integrated Risk Analysis

Outputs from computer vision and structured inputs can be used as indicators for the ML risk model, while NLP converts unstructured accident descriptions into structured information. The dashboard will clearly identify the result as a model-based risk estimate.

6. System Architecture

INPUT	AI PROCESSING	OUTPUT
Road Image / Video	YOLO / Computer Vision	Detected Objects + Counts
Traffic / Accident Data	ML Classification	Low / Medium / High Risk
Accident Description	NLP Extraction	Structured Accident Information
All Module Results	Integration + Dashboard	RoadSafe AI Analysis

7. Technologies

Area	Technologies
Programming	Python
Machine Learning	Scikit-learn, XGBoost, Pandas, NumPy
Deep Learning	YOLO, PyTorch
Computer Vision	OpenCV
NLP	spaCy, Hugging Face Transformers (where appropriate)
Visualization	Matplotlib, Plotly
Web Application	Streamlit

8. Dataset

The project will use publicly available datasets appropriate for each module. A traffic/accident dataset will support the ML risk model, a road-object dataset or pretrained model will support computer vision, and suitable accident-report text will support the NLP component. Dataset selection will consider relevance, quality, sample size, and licensing.

9. Model Evaluation

The ML classifier will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. Where appropriate, ROC-AUC may also be reported. The computer-vision component can be evaluated using precision, recall, IoU, and mAP where labeled evaluation data are available. NLP extraction will be checked against expected information and evaluated using suitable precision, recall, and F1 measures when labeled data are available.

10. User Interface

Screen	Function
Dashboard	Overview of traffic counts, risk result, recent analyses, and key statistics.
Traffic Vision	Upload road image, detect objects, display annotated result, and show counts.
Accident Risk	Enter structured road/environmental features and obtain a model-based risk category.
Accident Report	Enter a report and extract structured accident information.
Model Performance	Display evaluation metrics, confusion matrix, and charts.

11. Expected Results

• Detection of common road objects from uploaded images.

• Vehicle counts and an estimated traffic-density indicator.

• A trained ML model that provides a Low, Medium, or High risk classification.

• Extraction of important information from accident descriptions.

• Interactive charts and model-performance visualizations.

• A unified Streamlit application demonstrating ML, DL, CV, and NLP integration.

12. Scope

Included

Road-image object detection; vehicle counting; traffic-density estimation; ML-based risk classification; NLP accident-report analysis; visualization; Streamlit deployment; and model evaluation.

Future Scope

Real-time CCTV streams, live traffic APIs, GPS/map integration, live weather information, advanced vehicle tracking, lane-violation analysis, and edge-device deployment can be considered in future versions.

13. Limitations

Model performance depends on dataset quality and representativeness. Computer-vision accuracy may vary with lighting, weather, camera angle, and image quality. Public datasets may not fully represent local road conditions. The risk output is a model-based estimate and does not guarantee that an accident will or will not occur.

14. Seven-Day Development Plan

Day	Planned Work
Day 1	Dataset selection, preprocessing, EDA, initial ML model
Day 2	ML model comparison, evaluation, and model saving
Day 3	YOLO/OpenCV computer-vision module and object counting
Day 4	NLP accident-report analyzer
Day 5	Streamlit UI and integration of modules
Day 6	Testing, charts, evaluation, and improvements
Day 7	Documentation, presentation, final testing, and demo

15. Expected Learning Outcomes

The project will provide practical experience in machine learning, deep learning, object detection, computer vision, natural language processing, data preprocessing, feature engineering, model evaluation, visualization, AI-system integration, and Streamlit application development.

16. Conclusion

RoadSafe AI will demonstrate the practical integration of four major AI disciplines in a focused road-safety application. By combining computer-vision object detection, machine-learning risk classification, and NLP-based accident-report

analysis, the project will provide a complete academic prototype that can be implemented and demonstrated within a one-week development period.